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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

December 2024

ME1003-2 – ELEMENTS OF MECHANICAL ENGINEERING

(For ACT, BT, EC, EE, ME, VLSI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The slope of the stress-strain curve in the elastic deformation region is
 - A) Elastic modulus
 - B) Plastic modulus
 - C) Poisson's ratio
 - D) Bulk modulus
2. The heating of dry and saturated steam above saturation temperature is known as
 - A) Enthalpy
 - B) Superheating
 - C) Super saturation
 - D) Latent heat
3. The property which permits a material to be drawn out longitudinally to a reduced section under the action of tensile force
 - A) Ductility
 - B) Elasticity
 - C) Malleability
 - D) Plasticity
4. The mode of heat transfer that is generally observed in liquid medium is,
 - A) Conduction
 - B) Convection
 - C) Radiation
 - D) Both radiation conduction
5. Pelton turbine is suitable for
 - A) high head and low discharge
 - B) high head and high discharge
 - C) low head and high discharge
 - D) low head and low discharge
6. Centrifugal pumps are used to transport
 - A) Pressure
 - B) Speed
 - C) Power
 - D) Fluid
7. Function of an air compressor is to,
 - A) Discharge high pressure water
 - B) Suction of fuel in engines
 - C) Pressurize air and deliver
 - D) Compress the oil
8. In a gas turbine, which type of cycle is used where air is compressed and mixed with fuel before ignition?
 - A) Open cycle
 - B) Closed cycle
 - C) Rankine cycle
 - D) Otto cycle
9. The ratio of brake power to indicated power is called
 - A) Indicated thermal efficiency
 - B) Relative efficiency
 - C) Mechanical efficiency
 - D) Volumetric efficiency
10. The function of flywheel is to,
 - A) Reduce efficiency of engine
 - B) Store and release energy during work cycle
 - C) Regulate the speed
 - D) Increase weight of the engine
11. The expansion device in any refrigeration system is connected between
 - A) Evaporator and Condensor
 - B) Evaporator and Compressor
 - C) Compressor and Condensor
 - D) Evaporator and Absorber

12. How does a room air conditioner remove heat from the indoor air?
 A) By heating the air B) By compressing the air
 C) By evaporating a refrigerant D) By blowing cool outdoor air into the room
13. Worm gear is used, when two shafts are at
 A) Right angles to each other B) Some distance apart
 C) Radial direction D) 45 cm apart
14. Which joint formed by the following processes is the strongest.
 A) Brazing B) Soldering
 C) Gluing D) Welding
15. To obtain a higher torque, driven gear size should be,
 A) Smaller than driver B) Larger than driver
 C) Same as driver D) Do not depend on gear size
16. Soldering is having proportionate composition of,
 A) Lead and Copper B) Zinc and Tin
 C) Tin and Lead D) Zinc and Copper
17. The process of increasing the size of a already drilled hole is called
 A) Boring B) Reaming
 C) Counter sinking D) Facing
18. Soft automation is also called
 A) Fixed automation B) Integrated automation
 C) Flexible automation D) Programmable automation
19. Which part of a robot is typically responsible for end-effector tasks and tool manipulation?
 A) Controller B) Manipulator
 C) Sensor D) Actuator
20. Which lathe operation involves removing material from the end of a workpiece to make it flat and perpendicular to the axis?
 A) Turning B) Facing
 C) Knurling D) Thread cutting

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List and define any four mechanical properties of engineering materials.	08	L*1	1	1
	b) What is a compressor? Explain the working of single stage reciprocating air compressor with a neat sketch.	08	L2	2	1
2.	a) Explain the following: i. Conduction ii. Convection iii. Zeroth law of thermodynamics iv. First law of thermodynamics	08	L2	1	1
	b) Explain the working Principle of Closed cycle Gas turbine with a sketch.	08	L2	2	1
3.	a) Explain the working principle of Tidal power plant with a neat sketch.	08	L2	1	1
	b) Explain the working Principle of Impulse Steam Turbine (De Laval Turbine) with a sketch.	08	L2	2	1
Unit – II					
4.	a) Explain with figure the working principle of a 4-Stroke Diesel engine.	08	L2	3	1
	b) Explain the principle of MIG welding with suitable diagrams.	08	L2	4	1
5.	a) Describe Electric and Hybrid Vehicles with sketches.	08	L2	3	1
	b) Explain the process of brazing. Also mention their uses.	08	L2	4	1

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| 6. a) | Explain Parts of 4-stroke I.C. Engines with a sketch. | 08 | L2 | 3 | 1 |
| b) | In a crossed belt drive the difference in tensions between the tight and slack sides of the belt is 1000 N. Find the tension on the slack and tight sides. If the angle of contact is 160° and its coefficient of friction is 0.3. | 08 | L4 | 4 | 1 |

Unit – III

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|-------|--|----|----|---|---|
| 7. a) | Differentiate between Open and Closed loop control systems. | 08 | L2 | 5 | 1 |
| b) | Explain with figures the following Milling machine operations: | | | | |
| i. | Plane milling | 08 | L2 | 5 | 1 |
| ii. | Slot milling. | | | | |
| 8. a) | Explain with figures the following Drilling machine operations: | | | | |
| i. | Drilling | 08 | L2 | 5 | 1 |
| ii. | Boring | | | | |
| b) | Describe different Elements of automated systems with suitable sketches. | 08 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
